Additional Course Information – Not a Syllabus!

PHYS 675 – General Relativity
T Th 10:30-11:45, PHYS 234

Instructor: Prof. Paul Duffell
Office: PHYS 326
Office Hours: Tuesday 12:00-1:00pm
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Grader: Derek Ping

Course Description

An overview of Einstein's theory of General Relativity, covering the basics of non-Euclidean geometry, with applications in astrophysics.

Topics include: Special Relativity, Einstein four-vector formalism, Non-Euclidean metrics, Geodesics, Vector Transport, Curvature, Einstein Equation.

Applications include: Black holes, gravitational waves, cosmology.

Prerequisites:
PHYS 600 – Methods of Theoretical Physics I
(For ODEs & PDEs)

PHYS 610 – Advanced Theoretical Mechanics
(Lagrangian and Orbital Dynamics for computing geodesics)

PHYS 630 – Advanced Theory Of Electricity And Magnetism
(Probably the most important prerequisite, as E&M and GR are both classical field theories)

Textbooks

No Required Text. Some Nice Textbooks:
Carroll, S.M. *An Introduction to General Relativity, Spacetime and Geometry*
Hartle, J.B. *Gravity: An Introduction to Einstein's General Relativity*
Misner C.W., Thorne, K.S. & Wheeler, J.A. *Gravitation*

Grade System

Homework 50%
Participation 20%
Final Exam 30%

Course Outline

The course will consist of 14 weeks of instruction, roughly (and incompletely) outlined below. This outline will be modified based on student interests and time constraints:

Part I – Preliminaries	
Week 1 – Special Relativity and Einstein Summation	Carroll §1, Hartle §4
Week 2 – Four-Vectors and Relativistic Dynamics	Carroll §1, Hartle §5
Week 3 – Equivalence Principle and Gravity as Geometry	Hartle §6
Week 4 – "Curved" Metrics and Geodesics	Carroll §2, Hartle §7 & 8
Part II – Examples	
Week 5 – Schwarzchild Metric and Orbits around a Black Hole	Carroll §5, Hartle §9 & 12
Week 6 – Kerr Metric: Spinning Black Holes	Carroll §6, Hartle §14
Week 7 – Gravitational Waves	Carroll §7, Hartle §16
Week 8 – Cosmology and the Friedmann Equation	Carroll §8, Hartle §18
Part III – General Relavitivity	
Week 9 – Parallel Transport	Carroll §3, Hartle §20 & 21
Week 10 – Curvature and the Einstein Equation	Carroll §4, Hartle §21 & 22
Week 11 – Schwarzchild and Friedmann Revisited	Carroll §5 & 8, Hartle §24
Part IV – Extras	
Week 12 – Gravitational Wave Emission	Carroll §7, Hartle §23
Week 13 – What's a Vector? Manifolds and Math	Carroll §2 & 3, Hartle §20
Week 14 – Einstein as an Evolution Equation: 3+1 Spacetime Splitting	N/A

Additional Details

Homework will be assigned and turned in via Brightspace. Homework grades are mostly based on completion of all homework problems. Wrong answers are not significantly penalized so long as all problems are attempted. Students are encouraged to work together to solve problems, but your final writeup should be your own work. Solutions to homework problems will be presented in-class upon request.

The participation component will be judged with modest expectations, but it is asked that students be engaged in class. If everyone is reasonably engaged and responsive, then everyone gets full participation credit.

The final exam might happen. If it does not happen, we will re-balance homework and participation grades to 70% and 30%, respectively.

Outside of class, communication will largely be handled via email. In order to receive course information in a timely manner, please check your Purdue email regularly.